

Technical Documentation

Diesel Engine
12 V 2000 G60

Operating Instructions
M015395/10E



Das Handbuch ist zur Vermeidung von Störungen oder Schäden beim Betrieb zu beachten und daher vom Betreiber dem jeweiligen Wartungs- und Bedienungspersonal zur Verfügung zu stellen. Außerhalb dieses Verwendungszwecks darf das Handbuch ohne unsere vorherige Zustimmung nicht benutzt, vervielfältigt oder Dritten sonstwie zugänglich gemacht werden.

Änderungen bleiben vorbehalten.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation. Other than for this purpose, the handbook shall not be reproduced, used or disclosed to others without our prior consent.

Subject to alterations and amendments.

Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite. En dehors de cet usage, le manuel ne pourra être utilisé ni reproduit ou rendu accessible de quelque autre manière à des tiers, sans notre consentement préalable.

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El Manual debe tenerse presente para evitar anomalías o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio. Fuera de este fin de aplicación, el Manual no se debe utilizar, copiar ni poner en manos de terceros, sin nuestro consentimiento previo.

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta. Senza nostra approvazione preventiva non è ammesso impiegare il manuale per scopi diversi, riprodurlo o metterlo a disposizione di terzi.

Con riserva di modifiche.

Wichtig – Important – Importante

Bitte die Karte „Inbetriebnahmemeldung“ abtrennen und ausgefüllt an MTU Friedrichshafen GmbH zurücksenden.

Die Informationen der Inbetriebnahmemeldung sind Grundlage für den vertraglich vereinbarten Logistik-Support (Gewährleistung, Ersatzteile etc.).

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).

Veillez séparer la carte “Signalisation de mise en service” et la renvoyer à la MTU Friedrichshafen GmbH.

Les informations concernant la signalisation de mise en service constituent la base pour l’assistance en exploitation contractuelle (garantie, rechanges, etc.).

Rogamos separar la tarjeta “Aviso de puesta en servicio” y la devuelvan rellena a MTU Friedrichshafen GmbH.

Las informaciones respecto al aviso de puesta en servicio constituyen la base para el soporte logístico contractual (garantía, piezas de repuesto, etc.).

Staccare “Avviso di messa in servizio” e rispedirlo debitamente compilato alla MTU Friedrichshafen GmbH.

Le informazioni ivi registrate sono la base per il supporto logistico contrattuale (garanzia, ricambi, ecc.).



Postcard MTU Friedrichshafen GmbH Abteilung SCSD 88040 Friedrichshafen GERMANY
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Bitte in Blockschrift ausfüllen!
Please use block capitals!
Prière de remplir en lettres capitales!
¡A rellenar en letras de imprenta!
Scrivere in stampatello!



Motornr.:
Engine No.:
Nº du moteur:
Nº de motor:
Motore N.:

Auftragsnr.:
MTU works order No.:
Nº de commande:
Nº de pedido:
N. commessa:

Motortyp:
Engine model:
Type du moteur:
Tipo de motor:
Motore tipo:

Inbetriebnahmedatum:
Date put into operation:
Mise en service le:
Fecha de puesta en servicio:
Messa in servizio il:

Eingebaut in:
Installation site:
Lieu de montage:
Lugar de montaje:
Installato:

Schiffstyp / Schiffshersteller:
Vessel/type/class / Shipyard:
Type du bateau / Constructeur:
Tipo de buque / Constructor:
Tipo di Barca / Costruttore:

Endabnehmer/Anschrift:
End user's address:
Adresse du client final:
Dirección del cliente final:
Indirizzo del cliente finale:

Bemerkung:
Remarks:
Remarques:
Observaciones:
Commento:

Inbetriebnahmemeldung

Commissioning Note

Notice de mise en service

Aviso de puesta en servicio

Avviso di messa in servizio

- !** **Safety Instructions**
Summary
- A** **Product Summary**
Engine layout, technical data
- C** **Operating Instructions**
Task descriptions
- D** **Maintenance Schedule**
Time schedule and maintenance task chart
- E** **Troubleshooting**
Causes, remedies
- F** **Drawings**
Engine cutaway drawings
- G** **Task Description**
Inspection and maintenance tasks
- K** **Preservation**
Engine preservation
- ?** **Keyword Index**
Alphabetical index of key terms
- Z** **Appendix**
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These Safety Instructions must be read and followed by any persons operating or carrying out maintenance on the machinery plant.

!000.000.02 Basic Safety Instructions

This DDC/MTU engine is a state-of-the-art product and conforms with all applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of

- Incorrect use
- Operation by unqualified personnel
- Modifications or conversions
- Non-compliance with the Safety Instructions

!000.000.03 Correct Use

The engine is intended solely for the use specified in the contract. Any other use is considered improper use. The manufacturer will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Correct use also includes observation of the operating instructions and compliance with the maintenance schedule (Section D).

!000.000.05 Personnel Requirements

Only fully trained or qualified personnel must be employed.

The specified legal minimum age must be respected.

!000.000.07 Modifications or Conversions

Modifications made by the customer to the engine may affect safety. No modifications or conversions must be implemented without prior consent from DDC/MTU.

No liability will be accepted for damage resulting from unauthorised modifications.

!000.000.09 Organisational Measures

The Operating Instructions must be made available to the operating personnel. It must be kept at hand near the engine and accessible to the operating and maintenance personnel at all times.

The personnel must be instructed on engine operation by means of the Operating Instructions, and in particular the safety instructions must be explained. This is especially important for personnel who work on the engine only on an occasional basis.

In addition to the Operating Instructions, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed.

!000.000.11 Auxiliary Materials

Fluids and lubricants must be kept in properly designated containers. When using fluids, lubricants and other chemical substances, follow the safety instructions applicable to the product. Take special care when using hot and caustic materials. When using inflammable materials, avoid all sparks and do not smoke.

Always use DDC/MTU-approved and tested fluids and lubricants – see current DDC/MTU Fluids and Lubricants Specification.

!000.000.13 Spare Parts

Spare parts must at least satisfy the requirements specified by the manufacturer. This is guaranteed when original components are used.

!000.000.15 Work Clothing

Work clothing must be tight-fitting so that it does not catch on rotating or projecting components. Jewellery must not be worn.

!000.000.17 Transportation

Lift the engine only with the lifting eyes provided. Use only the transport and lifting equipment approved by DDC/MTU. The engine must only be transported in installation position.

!000.000.19 Engine Operation

When the engine is running, ear protectors must always be worn in the engine room.

The engine room must be well ventilated.

Leaked or spilt engine oil and coolant must be cleaned up immediately or absorbed with appropriate bonding agents to avoid injuries caused by falling.

Exhaust gases from combustion engines are poisonous and injurious to health if inhaled. Inhalation of poisonous exhaust gases is a health hazard. The exhaust pipework must be leak-free and discharge exhaust gases to atmosphere.

During engine operation, do not touch battery terminals, generator terminals or cables. Inadequate protection of electrical components can lead to electric shocks and serious injuries.

!000.000.21 Maintenance and Repair

Compliance with maintenance specifications is an important safety factor.

Unless expressly permitted, no maintenance work must be carried out with the engine running. The engine must be secured against inadvertent starting and the battery disconnected. Persons not involved must keep clear.

Maintenance work must only be carried out by authorised, qualified personnel.

Use only tools in perfect condition.

Before starting work, relieve pressure in systems and H.P. lines which are to be opened.

Until the engine has cooled, the breather cap must not be opened. Release the breather cap and allow pressure to escape before cap removal.

Take special care when draining hot fluids – risk of injury. Drain the fluids into a suitable container and wipe up any spillages or soak up suitable bonding agent.

When changing engine oil or working on the fuel system, ensure that the engine room is adequately ventilated.

When carrying out welding work on the engine, it is essential to observe the relevant safety measures.

When working with compressed air, always wear protective goggles. The pressure must not exceed 3 bar. When working with liquid nitrogen, always wear protective clothing – risk of injury.

When working high on the engine, always use suitable ladders and work platforms. Make sure components are placed on stable surfaces.

Gases released from the battery are explosive. Avoid sparks and naked flames. Do not allow battery acids to come into contact with skin or clothing. Wear protective goggles. Do not place tools on the battery. Before connecting the cable to the battery, check battery polarity. Battery pole reversal may lead to injury through the sudden discharge of acid or bursting of the battery body.

Before barring the engine, make sure that nobody is standing in the danger zone. After working on the engine, check that all guards have been reinstalled and that all tools and loose components have been removed from the engine.

!000.000.23 Environmental Protection

Dispose of used fluids and lubricants and filters in accordance with local regulations.

Changes to the injection or control system can influence the engine performance and exhaust emissions. As a result, compliance with environmental regulations may no longer be guaranteed.